Separation of Variables

The Wave Equation on a Finite Interval

Prof. A. K. B. Chand

IIT Madras



The Idea of Separation of Variables

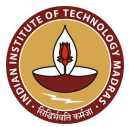
- We look for solutions of a linear PDE in product form

$$u(x, y) = X(x)\, Y(y).$$

- Each factor depends on one variable only.
- Then the partial derivatives are

$$u_x = X'Y, \quad u_{xx} = X''Y, \quad u_y = XY', \quad u_{yy} = XY'', \quad u_{xy} = X'Y'.$$

- Substitute these in the PDE and try to collect all x-terms on one side and all y-terms on the other.
- If this is possible, the PDE splits into two ordinary differential equations.
- Solve them to get $X(x)$ and $Y(y)$.
- The general solution is obtained by adding or integrating all such solutions (principle of superposition).



The Separation Constant

- Suppose the PDE can be written as

$$\frac{(\text{expression in } x \text{ only})}{X} = \frac{(\text{expression in } y \text{ only})}{Y}.$$

- The left side depends only on x; the right side only on y.
- Fix y and vary x: the left side cannot change. Fix x and vary y: the right side cannot change.
- Hence both sides are equal to the same constant μ, the separation constant.
- μ is arbitrary; it can also be written as $\mu = f(p)$ for a parameter $p \in \mathbb{R}$.
- In second order problems we consider the three cases $\mu = p^2 > 0$, $\mu = 0$, $\mu = -p^2 < 0$.
- Boundary conditions later select only certain values of μ (eigenvalues).



Principle of Superposition

- Let the PDE be linear and homogeneous, $\mathcal{L}[u] = 0$.
- If $u_1, u_2, \ldots, u_n$ are solutions, then so is $\sum_j c_j\, u_j$ for constants c_j.
- Separated solutions usually come as a family $u(x, y; \mu)$, one for each admissible μ.
- If μ takes discrete values $\mu_1, \mu_2, \ldots$:

$$u(x, y) = \sum_n C(\mu_n)\, u(x, y; \mu_n).$$

- If μ varies continuously:

$$u(x, y) = \int C(\mu)\, u(x, y; \mu)\, d\mu.$$

- For infinite sums and integrals, convergence and term by term differentiation must be justified.
- In boundary value problems, the coefficients $C(\mu_n)$ are fixed by the initial data.



Example 1: $u_x - u_y = 0$

- Let $u(x, y) = X(x)Y(y)$. Then $u_x = X'Y$, $u_y = XY'$.
- The PDE gives $X'Y - XY' = 0$, i.e.

$$\frac{X'}{X} = \frac{Y'}{Y} = \mu \quad \text{(say)}, \ \mu \in \mathbb{R}.$$

- $\displaystyle\int \frac{X'}{X}\,dx = \int \mu\,dx \Rightarrow \ln X - \ln c_1 = \mu x \Rightarrow X = c_1 e^{\mu x}$.
- Similarly $\displaystyle\int \frac{Y'}{Y}\,dy = \int \mu\,dy \Rightarrow Y = c_2 e^{\mu y}$.
- Hence $u(x, y) = X(x)Y(y) = c_3(\mu)\, e^{\mu(x+y)}$, where $c_3 = c_1 c_2$.



Example 1 (contd.): General Solution

- By superposition, the general solution is

$$u(x,y) = \int_{\mu} c_3(\mu)\, e^{\mu(x+y)}\, d\mu \quad (\mu \text{ continuous}),$$

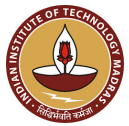
$$u(x,y) = \sum_{\mu} c_3(\mu)\, e^{\mu(x+y)} \quad (\mu \text{ discrete}).$$

- In both cases u is a function of $x+y$ only.
- This agrees with Lagrange's method: $u_x - u_y = 0$ gives $u = \phi(x+y)$.
- Separation produces the building blocks $e^{\mu(x+y)}$ of the arbitrary function ϕ.
- Particular solution with $u(x,0) = 3e^{2x}$: take only $\mu = 2$, $c_3 = 3$, so $u = 3e^{2(x+y)}$.



Example 2: $u_x + u_y = 0$

- Let $u = X(x)Y(y)$: $X'Y + XY' = 0$.
- Separate:
$$\frac{X'}{X} = -\frac{Y'}{Y} = \mu.$$
- Hence $X = c_1 e^{\mu x}$ and $Y = c_2 e^{-\mu y}$.
- Separated solutions: $u = c_3(\mu)\, e^{\mu(x-y)}$.
- General solution:
$$u(x,y) = \int_\mu c_3(\mu)\, e^{\mu(x-y)}\, d\mu \quad \text{or} \quad \sum_\mu c_3(\mu)\, e^{\mu(x-y)}.$$
- Again u depends only on $x - y$, the characteristic variable.



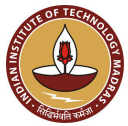
Example 3: $u_{xx} - u_y = 0$

- Let $u = X(x)Y(y)$: $u_{xx} = X''Y$, $u_y = XY'$.
- The PDE gives $X''Y - XY' = 0$, i.e.

$$\frac{X''}{X} = \frac{Y'}{Y} = \mu. \tag{1}$$

- Case $\mu = p^2 > 0$: $X'' - p^2 X = 0$, auxiliary equation $m^2 - p^2 = 0$, $m = \pm p$.
- $X = c_1 e^{px} + c_2 e^{-px}$ and $Y' = p^2 Y \Rightarrow Y = c_3 e^{p^2 y}$.
- Hence

$$u(x,y) = \left(c_4(p)\, e^{px} + c_5(p)\, e^{-px}\right) e^{p^2 y}, \quad c_4 = c_1 c_3,\ c_5 = c_2 c_3. \tag{2}$$



Example 3 (contd.): Cases $\mu = 0$ and $\mu < 0$

- Case $\mu = 0$: $X'' = 0 \Rightarrow X = Ax + B$; $Y' = 0 \Rightarrow Y = D$.

$$u(x, y) = Ex + F, \quad E = AD, \; F = BD. \tag{3}$$

- Case $\mu = -p^2 < 0$: $X'' + p^2 X = 0$, $m = \pm ip$.
- $X = c_6 \cos px + c_7 \sin px$ and $Y' = -p^2 Y \Rightarrow Y = c_8 e^{-p^2 y}$.

$$u(x, y) = \left(c_9(p) \cos px + c_{10}(p) \sin px\right) e^{-p^2 y}. \tag{4}$$

- The case $\mu < 0$ gives solutions that decay in y; it is the one used for heat flow.



Example 3 (contd.): General Solution

- Adding the solutions (2)–(4) over all admissible p:
- If p varies continuously,

$$u = Ex+F+\int_p \left(c_4 e^{px}+c_5 e^{-px}\right) e^{p^2 y}\, dp+\int_p \left(c_9 \cos px+c_{10} \sin px\right) e^{-p^2 y}\, d$$

- If p is discrete,

$$u = Ex+F+\sum_p \left(c_4 e^{px}+c_5 e^{-px}\right) e^{p^2 y}+\sum_p \left(c_9 \cos px+c_{10} \sin px\right) e^{-p^2 y}.$$

- Here c_4, c_5, c_9, c_{10} are functions of p.
- The PDE $u_y = u_{xx}$ is the heat equation with y as time.



Example 4: $u_t - u_{xx} - 2u_x = 0$

- Let $u(x,t) = X(x)T(t)$: $u_t = XT'$, $u_x = X'T$, $u_{xx} = X''T$.
- The PDE gives $XT' - X''T - 2X'T = 0$, so

$$\frac{T'}{T} = \frac{X'' + 2X'}{X} = \mu. \tag{1}$$

- The X-equation is $X'' + 2X' - \mu X = 0$ with auxiliary equation $m^2 + 2m - \mu = 0$:

$$m = -1 \pm \sqrt{1+\mu}. \tag{2}$$

- Three cases: $\mu > -1$, $\mu = -1$, $\mu < -1$.
- The critical value is now $\mu = -1$, not $\mu = 0$, because of the term $2u_x$.



Example 4 (contd.): The Three Cases

- Case $\mu > -1$: put $\mu = -1 + p^2$, $p > 0$. Then $m = -1 \pm p$ and $T = c_3 e^{(p^2-1)t}$.

$$u = \left[c_4(p)\, e^{(-1+p)x} + c_5(p)\, e^{-(1+p)x}\right] e^{(p^2-1)t}. \tag{3}$$

- Case $\mu = -1$: $m = -1$ (twice): $X = (c_6 + c_7 x)e^{-x}$, $T = c_8 e^{-t}$.

$$u = (c_9 + c_{10}x)\, e^{-(x+t)}. \tag{4}$$

- Case $\mu < -1$: put $\mu = -1 - p^2$. Then $m = -1 \pm ip$ and $T = c_{13} e^{-(1+p^2)t}$.

$$u = e^{-[x+(1+p^2)t]}\big(c_{14}(p)\cos px + c_{15}(p)\sin px\big). \tag{5}$$



Example 4 (contd.): General Solution

- Using (3)–(5), for continuous p:

$$u(x,t) = (c_9 + c_{10}x)e^{-(x+t)} + \int_p \left[c_4 e^{(-1+p)x} + c_5 e^{-(1+p)x}\right] e^{(p^2-1)t}\, dp$$
$$+ \int_p \left(c_{14}\cos px + c_{15}\sin px\right) e^{-[x+(1+p^2)t]}\, dp.$$

- For discrete p the integrals are replaced by sums $\sum_p$.
- Remark: the substitution $u = e^{-x-t}v$ reduces the PDE to $v_t = v_{xx}$.
- This explains the common factor e^{-x} and the shift of the critical value to $\mu = -1$.



Example 5: $u_{xy} = e^{-y}$, $u(x,0) = 0$, $u_y(0,y) = 0$

- Let $u = X(x)Y(y)$: then $u_{xy} = X'Y' = e^{-y}$, so

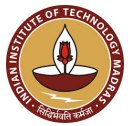
$$X' = \frac{e^{-y}}{Y'} = \mu \quad (\mu \neq 0).$$

- $X' = \mu \Rightarrow X = \mu x + c_1.$ (1)
- $Y' = \frac{1}{\mu} e^{-y} \Rightarrow Y = -\frac{1}{\mu} e^{-y} + c_2.$ (2)
- $u(x,0) = X(x)Y(0) = 0 \Rightarrow Y(0) = 0$ (else $u \equiv 0$), so $c_2 = \frac{1}{\mu}$ and $Y = \frac{1 - e^{-y}}{\mu}$.
- Note $Y'(y) \neq 0$. Then $u_y(0,y) = X(0)Y'(y) = 0 \Rightarrow X(0) = 0 \Rightarrow c_1 = 0.$
- Hence $u(x,y) = \mu x \cdot \frac{1 - e^{-y}}{\mu} = x(1 - e^{-y}).$
- Check: $u_{xy} = e^{-y}$, $u(x,0) = 0$, $u_y(0,y) = 0$.



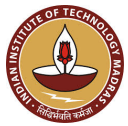
When Does Separation Work?

- The method needs the PDE to split after dividing by XY.
- It works for $a(x)u_{xx} + b(x)u_x + c(y)u_{yy} + d(y)u_y = 0$-type equations.
- It may fail when mixed terms like u_{xy} appear together with u_{xx} or u_{yy}.
- Example: $u_{xx} + u_{xy} + u_{yy} = 0$ gives $X''Y + X'Y' + XY''$, which does not separate directly.
- One may first change variables (e.g. to canonical form) and then separate.
- The boundary must also fit the coordinates: lines $x = \text{const}$ and $y = \text{const}$.
- For finite intervals, boundary conditions lead to eigenvalue problems (next sections).



Homework: Separation of Variables

- Solve by the method of separation of variables:
- (i) $2x\,u_x - 3y\,u_y = 0$.
- (ii) $y^2 u_x + x^2 u_y = 0$.
- (iii) $u_x + 4u_y = 0, \;\; u(0,y) = 8e^{-3y}$.
- (iv) $4u_x + u_y = 3u, \;\; u(0,y) = 2e^{3y} + e^{-5y}$.
- (v) $u_x = 2u_y + u, \;\; u(x,0) = 6e^{-3x} + 5e^{2x} + e^{5x}$.
- (vi) $u_{xx} = u_y + 2u, \;\; u(0,y) = 0, \;\; u_x(0,y) = 1 + e^{-3y}$.
- Hint for (iv): each exponential in the data gives one value of μ; add the separated solutions.



Orthogonality Relations on $[0, L]$

- For positive integers m, n:

$$\int_0^L \sin\frac{n\pi x}{L}\,\sin\frac{m\pi x}{L}\,dx = \begin{cases} 0, & m \neq n, \\ \dfrac{L}{2}, & m = n. \end{cases}.$$

- For non-negative integers m, n:

$$\int_0^L \cos\frac{n\pi x}{L}\,\cos\frac{m\pi x}{L}\,dx = \begin{cases} 0, & m \neq n, \\ \dfrac{L}{2}, & m = n \neq 0, \\ L, & m = n = 0. \end{cases}.$$

- Also $\displaystyle\int_{-L}^{L} \sin\frac{n\pi x}{L}\cos\frac{m\pi x}{L}\,dx = 0$ for all m, n (odd integrand).
- These relations let us pick out one coefficient at a time from an infinite series.



Fourier Sine and Cosine Series

- Sine series: a piecewise smooth f on $[0, L]$ can be expanded as

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}, \qquad b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L}\, dx.$$

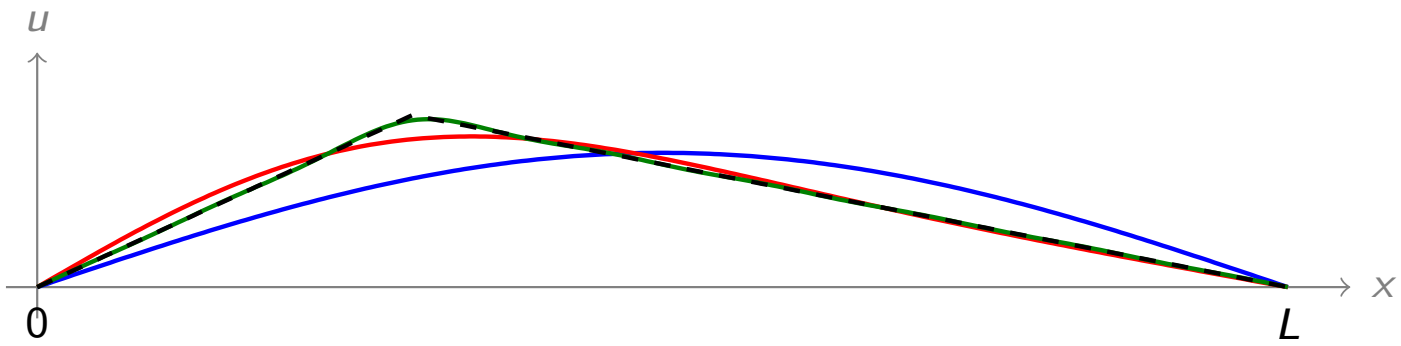
- It is the Fourier series of the odd $2L$-periodic extension of f.
- Cosine series:

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L}, \quad a_0 = \frac{1}{L} \int_0^L f\, dx, \quad a_n = \frac{2}{L} \int_0^L f \cos \frac{n\pi x}{L}\, dx$$

- It is the Fourier series of the even $2L$-periodic extension of f.
- At a jump, the series converges to the average of the one-sided limits.
- Derivation: multiply by $\sin \frac{m\pi x}{L}$ (or cos), integrate over $[0, L]$, use orthogonality.



Picture: Partial Sums of a Sine Series



- Triangular profile (dashed) and the partial sums with 1 (blue), 3 (red) and 15 (green) terms.
- The coefficients decay like $1/n^2$, so the convergence is fast and uniform.

Vibrating String with Fixed Ends

- A string of length L is fixed at $x = 0$ and $x = L$:

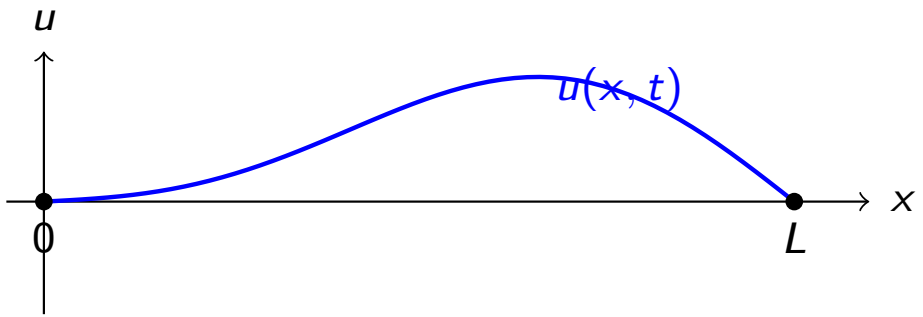
$$u_{tt} = k^2 u_{xx}, \qquad 0 < x < L, \ t > 0. \tag{1}$$

- Boundary conditions (fixed ends, Dirichlet):

$$u(0, t) = 0, \qquad u(L, t) = 0, \qquad t \geq 0. \tag{2}$$

- Initial conditions:

$$u(x, 0) = f(x) \text{ (initial deflection)}, \qquad u_t(x, 0) = g(x) \text{ (initial velocity)}. \tag{3}$$



Steps of the Method

- Step 1: Let $u(x,t) = X(x)T(t)$ and find the corresponding ODEs.
- Step 2: Transfer the boundary conditions to X.
- Step 3: Solve the resulting eigenvalue problem (Sturm–Liouville problem) to get $X_n(x)$.
- Step 4: Use the eigenvalues to compute $T_n(t)$.
- Step 5: Form $u_n(x,t) = X_n(x)T_n(t)$ and $u(x,t) = \sum_n u_n(x,t)$.
- Step 6: Compute the coefficients from the initial conditions using Fourier series.
- The same steps apply to free and mixed boundary conditions.



Separating the Wave Equation

- Let $u(x,t) = X(x)T(t)$. Then $u_{tt} = XT''$, $u_{xx} = X''T$.
- From (1): $XT'' = k^2 X''T$, so

$$\frac{X''}{X} = \frac{1}{k^2}\frac{T''}{T} = \mu \quad \text{(say)}.$$

- This gives two ODEs:

$$X'' - \mu X = 0, \tag{4}$$

$$T'' - \mu k^2\, T = 0. \tag{5}$$

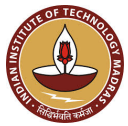
- From (2): $X(0)T(t) = 0$ and $X(L)T(t) = 0$ for all t.
- If $T \equiv 0$, then $u \equiv 0$, which is of no interest. Hence

$$X(0) = 0, \qquad X(L) = 0. \tag{6}$$



The Eigenvalue Problem: $\mu \geq 0$

- Consider the Sturm–Liouville problem (4), (6): $X'' - \mu X = 0$, $X(0) = X(L) = 0$.
- Case I: $\mu = 0$. $X'' = 0 \Rightarrow X = c_1 x + c_2$.
- $X(0) = 0 \Rightarrow c_2 = 0$; $X(L) = c_1 L = 0 \Rightarrow c_1 = 0$.
- So $X \equiv 0$: no non-trivial solution.
- Case II: $\mu = p^2 > 0$. $m = \pm p$, $X = c_3 e^{-px} + c_4 e^{px}$.
- $X(0) = c_3 + c_4 = 0 \Rightarrow c_3 = -c_4$.
- $X(L) = c_4 \left(e^{pL} - e^{-pL}\right) = 0 \Rightarrow c_4 = 0$, since $e^{pL} \neq e^{-pL}$ for $p > 0$.
- So again $X \equiv 0$. Non-negative μ give only the trivial solution.



The Eigenvalue Problem: $\mu < 0$

- Case III: $\mu = -p^2 < 0$, $p > 0$. $m = \pm ip$:

$$X(x) = A\cos px + B\sin px.$$

- $X(0) = 0 \Rightarrow A = 0$.
- $X(L) = 0 \Rightarrow B\sin pL = 0$.
- For a non-trivial solution $B \neq 0$, so $\sin pL = 0 = \sin n\pi$:

$$\boxed{p = \frac{n\pi}{L}, \quad n \in \mathbb{N}.}.$$

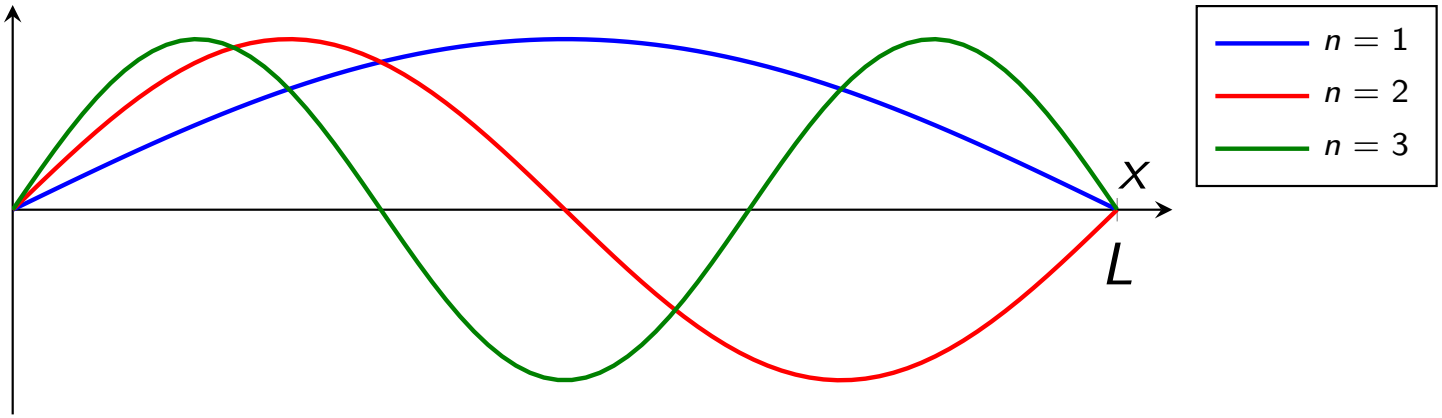
- Taking $B = 1$, the eigenfunctions are

$$X_n(x) = \sin\frac{n\pi x}{L}, \qquad n = 1, 2, 3, \ldots.$$

- The eigenvalues are $\mu_n = -\left(\frac{n\pi}{L}\right)^2$.



Picture: The First Normal Modes (Fixed Ends)



- $X_n(x) = \sin(n\pi x/L)$ has $n-1$ interior zeros (nodes) at $x = jL/n$.
- Each eigenfunction vanishes at both ends, as required by the fixed end conditions.



The Time Factor and Product Solutions

- With $\mu = -\left(\frac{n\pi}{L}\right)^2$, equation (5) becomes

$$T'' + \lambda_n^2 T = 0, \qquad \lambda_n = \frac{n\pi k}{L}. \tag{7}$$

- Auxiliary equation $m^2 + \lambda_n^2 = 0$, $m = \pm i\lambda_n$, so

$$T_n(t) = B_n \cos \lambda_n t + B_n^* \sin \lambda_n t, \qquad n \in \mathbb{N}.$$

- Hence the solutions of (1) satisfying (2) are

$$u_n(x,t) = X_n(x)T_n(t) = \left(B_n \cos \lambda_n t + B_n^* \sin \lambda_n t\right) \sin \frac{n\pi x}{L}.$$

- Each u_n is called the n-th normal mode of the string.



Superposition: The Series Solution

- By the principle of superposition, the general solution of (1)–(2) is

$$\boxed{u(x,t) = \sum_{n=1}^{\infty} \left(B_n \cos \lambda_n t + B_n^* \sin \lambda_n t \right) \sin \frac{n\pi x}{L}, \quad \lambda_n = \frac{n\pi k}{L}.} \tag{8}$$

- Each term satisfies the PDE and both boundary conditions.
- It remains to choose B_n and B_n^* so that the initial conditions (3) hold.
- This requires expanding f and g in Fourier sine series on $[0, L]$.



Determining B_n from the Initial Deflection

- Put $t = 0$ in (8) and use $u(x, 0) = f(x)$:

$$\sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L} = f(x).$$

- This is the Fourier sine series of f on $[0, L]$. By orthogonality,

$$\boxed{B_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} \, dx, \quad n \in \mathbb{N}.} \tag{9}$$

- Derivation: multiply by $\sin \frac{m\pi x}{L}$ and integrate; only the term $n = m$ survives.
- B_n is the amplitude of the n-th mode due to the initial shape.



Determining B_n^* from the Initial Velocity

- Differentiate (8) term by term w.r.t. t:

$$u_t(x,t) = \sum_{n=1}^{\infty} \lambda_n \big(- B_n \sin \lambda_n t + B_n^* \cos \lambda_n t \big) \sin \frac{n\pi x}{L}.$$

- Put $t = 0$ and use $u_t(x,0) = g(x)$:

$$\sum_{n=1}^{\infty} \lambda_n B_n^* \sin \frac{n\pi x}{L} = g(x) \;\Rightarrow\; \lambda_n B_n^* = \frac{2}{L} \int_0^L g(x) \sin \frac{n\pi x}{L}\, dx.$$

- Since $\lambda_n = n\pi k/L$:

$$\boxed{B_n^* = \frac{2}{n\pi k} \int_0^L g(x) \sin \frac{n\pi x}{L}\, dx, \quad n \in \mathbb{N}.} \qquad (10)$$

- Substituting (9) and (10) in (8) gives the solution of the problem.



Special Cases

- Zero initial deflection ($f \equiv 0$): then $B_n = 0$ for all n and

$$u(x,t) = \sum_{n=1}^{\infty} B_n^* \sin \lambda_n t \, \sin \frac{n\pi x}{L}, \quad B_n^* \text{ from (10)}.$$

- Zero initial velocity ($g \equiv 0$): then $B_n^* = 0$ for all n and

$$u(x,t) = \sum_{n=1}^{\infty} B_n \cos \lambda_n t \, \sin \frac{n\pi x}{L}, \quad B_n \text{ from (9)}.$$

- In general the solution is the sum of these two solutions (linearity).
- If f is a finite sum of sine modes, the solution is a finite sum: no integration needed.
- Example: $f = \sin \frac{\pi x}{L} + \frac{1}{2} \sin \frac{3\pi x}{L}$, $g = 0$ gives $u = \cos \lambda_1 t \sin \frac{\pi x}{L} + \frac{1}{2} \cos \lambda_3 t \sin \frac{3\pi x}{L}$.

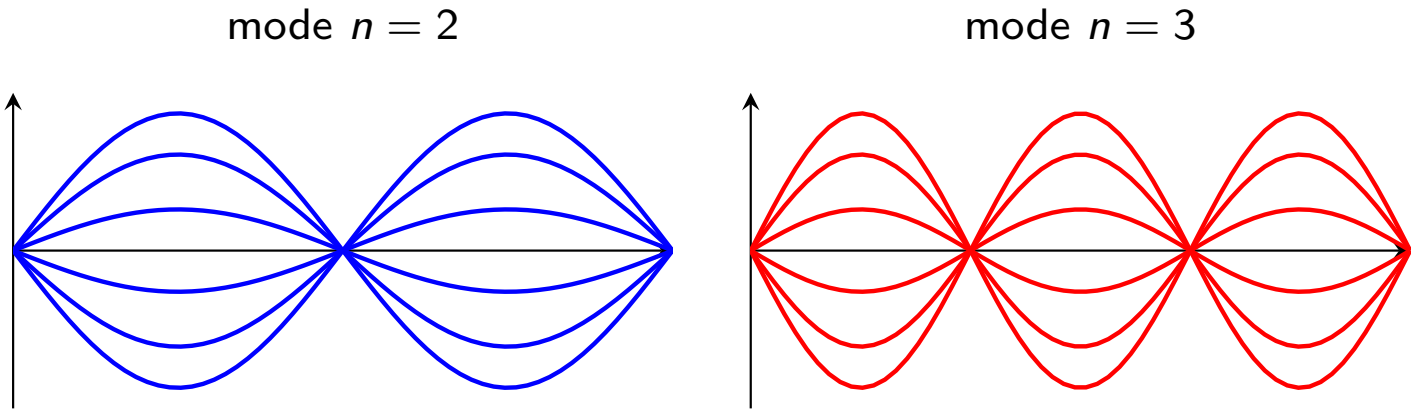


Eigenvalues, Spectrum and Frequencies

- The numbers $\lambda_n = \dfrac{n\pi k}{L}$, $n \in \mathbb{N}$, are called the eigenvalues or characteristic values.
- The set $\{\lambda_1, \lambda_2, \lambda_3, \dots\}$ is the spectrum of the vibrating string.
- The n-th mode oscillates in time with angular frequency λ_n.
- Its frequency is $\nu_n = \dfrac{\lambda_n}{2\pi} = \dfrac{nk}{2L}$ cycles per unit time.
- $\nu_1 = \dfrac{k}{2L}$ is the fundamental frequency; $\nu_n = n\nu_1$ are the harmonics (overtones).
- For a string with tension τ and density ρ, $k = \sqrt{\tau/\rho}$, so $\nu_1 = \frac{1}{2L}\sqrt{\tau/\rho}$.
- Tighter or shorter strings vibrate at higher pitch: this is how string instruments are tuned.



Picture: Standing Waves



- Each normal mode $\cos \lambda_n t \sin \frac{n\pi x}{L}$ keeps its shape and only changes its amplitude: a standing wave.
- The nodes $x = jL/n$ never move; points between them oscillate up and down.



Link with D'Alembert's Solution

- Take $g = 0$. Using $2\sin A\cos B = \sin(A-B) + \sin(A+B)$:

$$\cos\frac{n\pi kt}{L}\sin\frac{n\pi x}{L} = \frac{1}{2}\left(\sin\frac{n\pi(x-kt)}{L} + \sin\frac{n\pi(x+kt)}{L}\right).$$

- Hence

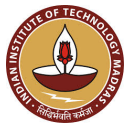
$$u(x,t) = \frac{1}{2}\sum_{n=1}^{\infty} B_n \sin\frac{n\pi(x-kt)}{L} + \frac{1}{2}\sum_{n=1}^{\infty} B_n \sin\frac{n\pi(x+kt)}{L}.$$

- Therefore

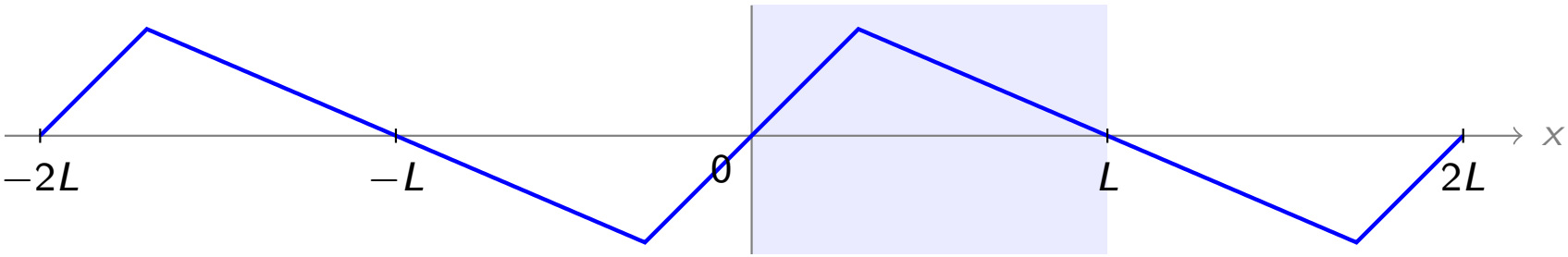
$$u(x,t) = \frac{1}{2}\left[f^*(x-kt) + f^*(x+kt)\right],$$

where f^* is the odd periodic extension of f with period $2L$.

- Every standing wave is a sum of two travelling waves moving in opposite directions.



Picture: Odd Periodic Extension f^*

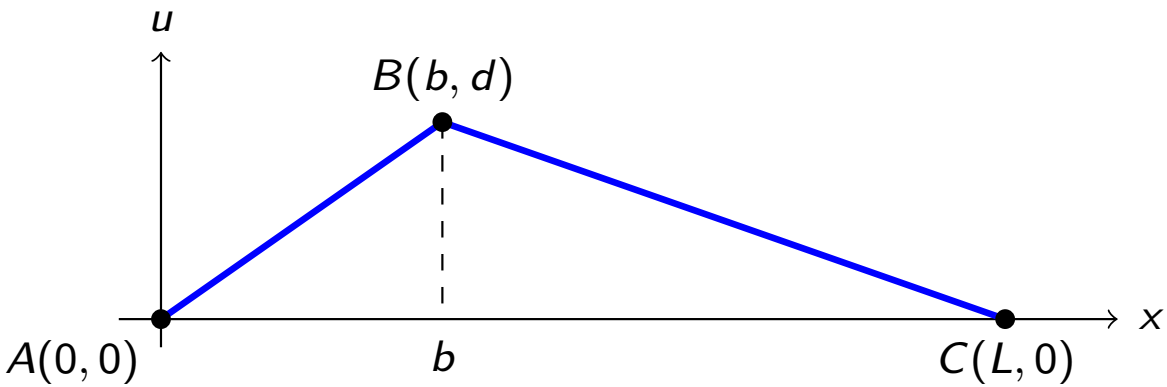


- f is given on $[0, L]$ (shaded); it is reflected oddly about $x = 0$ and repeated with period $2L$.
- The fixed ends $x = 0$ and $x = L$ become points of odd symmetry of f^*, so $u = 0$ there.
- This is the method of images of the semi-infinite string applied at both ends.



Example 6: The Plucked String

- A string of length L is fastened at both ends A and C.
- At a distance b from A the string is displaced transversely by a distance d and released from rest.
- Find the subsequent motion. (Initial deflection $\neq 0$, initial velocity $= 0$.).



- The initial deflection is given by the lines AB and BC:

$$f(x) = \begin{cases} \dfrac{d}{b}\,x, & 0 \le x \le b, \\ \dfrac{d\,(x-L)}{b-L}, & b \le x \le L. \end{cases}$$

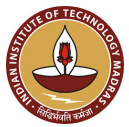
Example 6 (contd.): Computing B_n

- The problem: $u_{tt} = k^2 u_{xx}$, $u(0,t) = u(L,t) = 0$, $u(x,0) = f(x)$, $u_t(x,0) = 0$.
- Since $g = 0$: $u(x,t) = \sum_{n=1}^{\infty} B_n \cos\frac{n\pi kt}{L} \sin\frac{n\pi x}{L}$.
- $B_n = \dfrac{2}{L}\displaystyle\int_0^b \frac{d}{b}\, x \sin\frac{n\pi x}{L}\, dx + \frac{2}{L}\frac{d}{b-L}\int_b^L (x-L)\sin\frac{n\pi x}{L}\, dx.$
- Integrate by parts, using $\int x \sin\alpha x\, dx = -\frac{x\cos\alpha x}{\alpha} + \frac{\sin\alpha x}{\alpha^2}$:

$$B_n = \frac{2d}{bL}\Big[-\frac{Lx}{n\pi}\cos\frac{n\pi x}{L} + \frac{L^2}{n^2\pi^2}\sin\frac{n\pi x}{L}\Big]_0^b + \frac{2d}{L(b-L)}\Big[\cdots\Big]_b^L.$$

- The cosine terms cancel, leaving

$$B_n = \frac{2dL^2}{b(L-b)\, n^2\pi^2}\sin\frac{n\pi b}{L}.$$

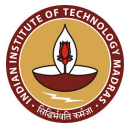


Example 6 (contd.): Solution

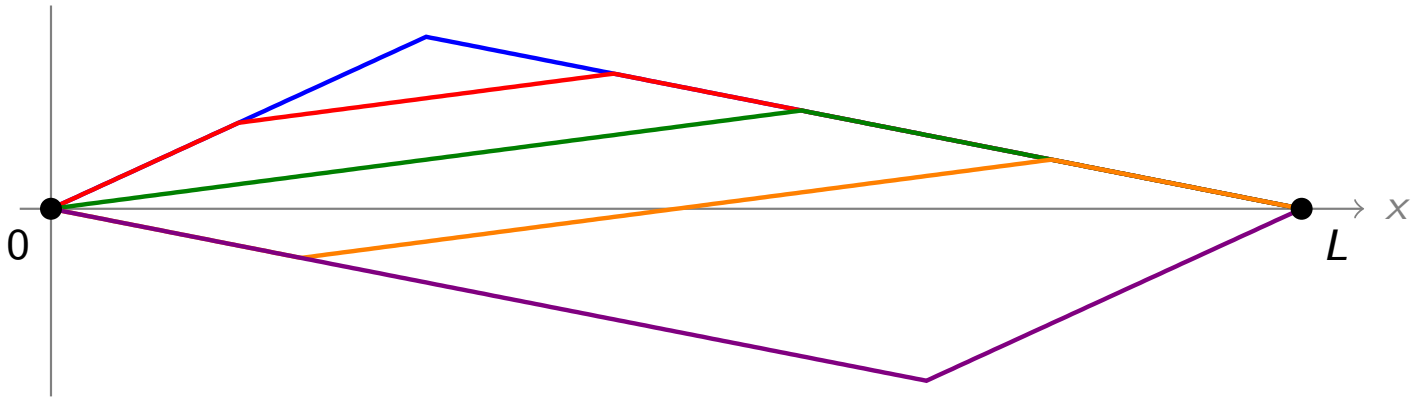
- Hence the displacement of the plucked string is

$$u(x,t) = \frac{2dL^2}{b(L-b)\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin\frac{n\pi b}{L} \sin\frac{n\pi x}{L} \cos\frac{n\pi kt}{L}.$$

- The amplitudes decay like $1/n^2$: higher harmonics are weak.
- If $b = L/m$, every m-th harmonic is missing, since $\sin(n\pi/m) = 0$ for $n = m, 2m, \ldots$.
- Plucking at the middle ($b = L/2$) excites only odd harmonics.
- The motion is periodic in t with period $2L/k$.
- By the D'Alembert form, u is piecewise linear in x at each instant.



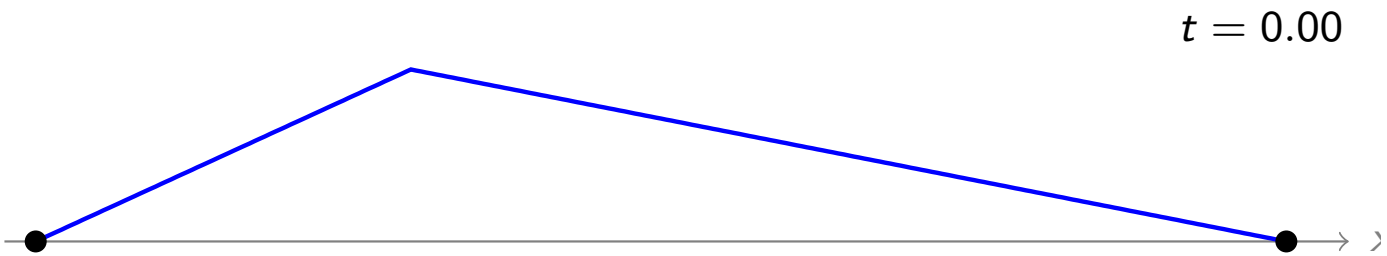
Picture: Snapshots of the Plucked String



- $b = 0.3L$, $k = 1$, $L = 4$: shapes at $t = 0$ (blue), 0.6, 1.2, 2 and $t = L/k$ (violet, inverted image).
- The corner splits into two corners moving with speed k and reflecting at the fixed ends.



Animation: Plucked String with Fixed Ends



- One full period $0 \leq t < 2L/k$; use the controls to pause or step (best viewed in Adobe Reader or Okular).
- At $t = L/k$ the string has the inverted, mirrored shape; at $t = 2L/k$ it returns to the start.



Example 7: String Struck from Rest

- A string is stretched between two fixed points at a distance 2ℓ apart.
- It is initially at rest in its equilibrium position, and is given the initial velocity

$$v(x) = \begin{cases} \dfrac{x}{\ell}, & 0 < x < \ell, \\ \dfrac{2\ell - x}{\ell}, & \ell < x < 2\ell. \end{cases}.$$

- (x is measured from one end.) Find the displacement.
- The problem: $u_{tt} = k^2 u_{xx}$, $u(0,t) = u(2\ell,t) = 0$, $u(x,0) = 0$, $u_t(x,0) = v(x)$.
- Here $L = 2\ell$ and $f = 0$, so

$$u(x,t) = \sum_{n=1}^{\infty} B_n^* \sin\frac{n\pi kt}{2\ell} \sin\frac{n\pi x}{2\ell}, \quad B_n^* = \frac{2}{n\pi k}\int_0^{2\ell} v(x)\sin\frac{n\pi x}{2\ell}\,dx.$$



Example 7 (contd.): Computing B_n^*

- Split the integral at $x = \ell$:

$$B_n^* = \frac{2}{n\pi k}\left\{ \int_0^{\ell} \frac{x}{\ell} \sin\frac{n\pi x}{2\ell}\, dx + \int_{\ell}^{2\ell} \frac{2\ell - x}{\ell} \sin\frac{n\pi x}{2\ell}\, dx \right\}.$$

- Integrating by parts, the terms with $\cos\frac{n\pi}{2}$ cancel and $\sin n\pi = 0$.
- Each integral contributes $\dfrac{4\ell}{n^2\pi^2}\sin\dfrac{n\pi}{2}$, so

$$B_n^* = \frac{2}{n\pi k} \cdot \frac{8\ell}{n^2\pi^2} \sin\frac{n\pi}{2} = \frac{16\ell}{k\pi^3} \cdot \frac{1}{n^3} \sin\frac{n\pi}{2}.$$

- Hence

$$u(x,t) = \frac{16\ell}{k\pi^3} \sum_{n=1}^{\infty} \frac{1}{n^3} \sin\frac{n\pi}{2} \sin\frac{n\pi kt}{2\ell} \sin\frac{n\pi x}{2\ell}.$$

- Only odd n contribute, with alternating signs $\sin\frac{n\pi}{2} = (-1)^{(n-1)/2}$.



Example 8: Non-zero Deflection and Velocity

- Solve $u_{tt} = k^2 u_{xx}$, $0 < x < L$, $t > 0$, with.
- (i) $u(0,t) = 0$, $u(L,t) = 0$;
- (ii) $u(x,0) = f(x) = \begin{cases} x, & 0 < x < L/2, \\ L - x, & L/2 < x < L; \end{cases}$.
- (iii) $u_t(x,0) = g(x) = x(x-L)$, $0 < x < L$.
- The solution has the general form (8) with $\lambda_n = n\pi k/L$:

$$u(x,t) = \sum_{n=1}^{\infty} \left(B_n \cos \frac{n\pi kt}{L} + B_n^* \sin \frac{n\pi kt}{L} \right) \sin \frac{n\pi x}{L}.$$

- By linearity, B_n comes from f and B_n^* comes from g independently.



Example 8 (contd.): The Coefficients

- From (9):

$$B_n = \frac{2}{L}\Big[\int_0^{L/2} x \sin\frac{n\pi x}{L}\,dx + \int_{L/2}^{L}(L-x)\sin\frac{n\pi x}{L}\,dx\Big] = \frac{4L}{n^2\pi^2}\sin\frac{n\pi}{2}.$$

- From (10), using $\int_0^L x(x-L)\sin\frac{n\pi x}{L}\,dx = -\frac{2L^3}{n^3\pi^3}\big(1-(-1)^n\big)$:

$$B_n^* = \frac{2}{n\pi k}\Big[-\frac{2L^3}{n^3\pi^3}\big(1-(-1)^n\big)\Big] = \begin{cases} -\dfrac{8L^3}{k\,n^4\pi^4}, & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}.$$

- The negative sign is natural: $g = x(x-L) \le 0$ pushes the string downward.



Example 8 (contd.): Solution

- Substituting in (8):

$$u(x,t) = \frac{4L}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{n^2}\sin\frac{n\pi}{2}\cos\frac{n\pi kt}{L}\sin\frac{n\pi x}{L} - \frac{8L^3}{k\pi^4}\sum_{n=1}^{\infty}\frac{1}{(2n-1)^4}\sin\frac{(2n-1)\pi kt}{L}\sin\frac{(2n-1)\pi x}{L}.$$

- The first sum is the motion due to the initial shape (released from rest).
- The second sum is the motion due to the initial velocity (from the equilibrium position).
- Both contain only odd harmonics, since f and g are symmetric about $x = L/2$.



Free Ends: Neumann Boundary Conditions

- If the ends are free to slide vertically, the slope vanishes there:

$$u_x(0,t) = 0, \qquad u_x(L,t) = 0, \qquad t \geq 0.$$

- With $u = XT$: $X'(0)T(t) = 0$, $X'(L)T(t) = 0$, so $X'(0) = 0 = X'(L)$.
- $\mu = p^2 > 0$: $X = c_3 e^{px} + c_4 e^{-px}$; $X'(0) = 0 \Rightarrow c_3 = c_4$; $X'(L) = 0 \Rightarrow c_3 = 0$. Trivial.
- $\mu = 0$: $X = c_1 x + c_2$, $X' = c_1 = 0$. So $X = c_2$ (a constant) is an eigenfunction with eigenvalue 0.
- $\mu = -p^2 < 0$: $X = A\cos px + B\sin px$; $X'(0) = Bp = 0 \Rightarrow B = 0$.
- $X'(L) = -Ap\sin pL = 0 \Rightarrow p = \dfrac{n\pi}{L}, \quad n \in \mathbb{N}$.
- Eigenfunctions: $X_n(x) = \cos\dfrac{n\pi x}{L}, \quad n = 0, 1, 2, \ldots$ (including the constant $n = 0$).



Free Ends: Series Solution

- For $n = 0$: $T'' = 0 \Rightarrow T_0 = A_0 + A_0^* t$ (a uniform drift of the whole string).
- For $n \geq 1$: $T_n = A_n \cos \lambda_n t + A_n^* \sin \lambda_n t$, $\lambda_n = n\pi k/L$.
- By superposition,

$$u(x,t) = A_0 + A_0^* t + \sum_{n=1}^{\infty} \left(A_n \cos \lambda_n t + A_n^* \sin \lambda_n t \right) \cos \frac{n\pi x}{L}.$$

- Initial conditions give Fourier cosine series:

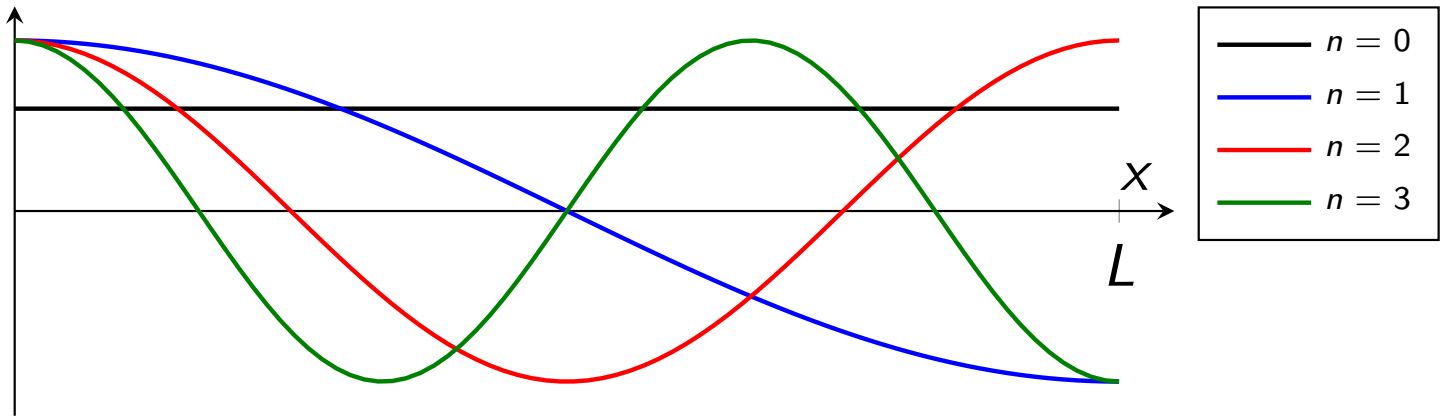
$$A_0 = \frac{1}{L} \int_0^L f \, dx, \quad A_n = \frac{2}{L} \int_0^L f \cos \frac{n\pi x}{L} \, dx,$$

$$A_0^* = \frac{1}{L} \int_0^L g \, dx, \quad A_n^* = \frac{2}{n\pi k} \int_0^L g \cos \frac{n\pi x}{L} \, dx.$$

- With $g = 0$: $u = \frac{1}{2}\left[f_e(x - kt) + f_e(x + kt) \right]$, $f_e =$ even $2L$-periodic extension of f.



Picture: Normal Modes with Free Ends



- $X_n = \cos(n\pi x/L)$: every mode has zero slope at both ends (horizontal tangents).
- The ends are antinodes: they move with maximal amplitude.



Example 9: $u_{tt} - 4u_{xx} = 0$ with Free Ends

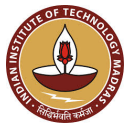
- Solve $u_{tt} - 4u_{xx} = 0$, $0 < x < \pi$, $t > 0$, with

  $u_x(0,t) = u_x(\pi,t) = 0, \qquad u(x,0) = 3\cos x, \qquad u_t(x,0) = 1 - \cos 4x.$

- Here $k = 2$, $L = \pi$, so $X_n = \cos nx$ and $\lambda_n = 2n$.
- General solution:

$$u(x,t) = A_0 + A_0^* t + \sum_{n=1}^{\infty} \left(A_n \cos 2nt + A_n^* \sin 2nt\right) \cos nx.$$

- $u(x,0) = A_0 + \sum A_n \cos nx = 3\cos x \Rightarrow A_0 = 0$, $A_1 = 3$, $A_n = 0$ for $n \neq 1$.
- $u_t(x,0) = A_0^* + \sum 2nA_n^* \cos nx = 1 - \cos 4x \Rightarrow A_0^* = 1$, $8A_4^* = -1$, others 0.

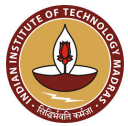


Example 9 (contd.): Solution

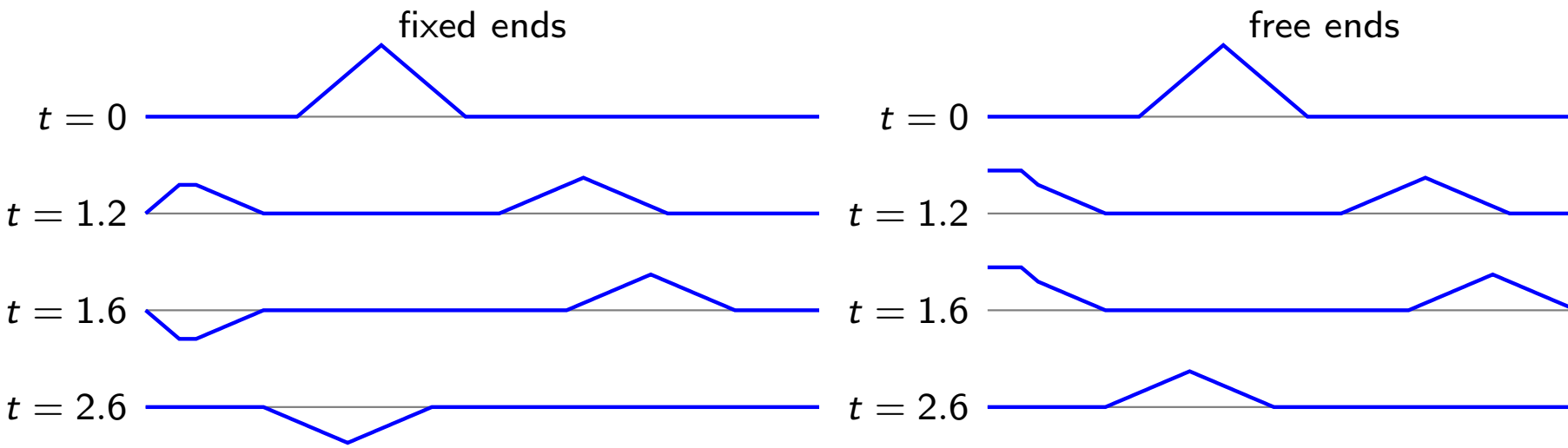
- Hence $A_4^* = -\frac{1}{8}$ and

$$u(x,t) = t + 3\cos 2t \cos x - \frac{1}{8}\sin 8t \cos 4x.$$

- Check the PDE: $u_{tt} = -12\cos 2t \cos x + 8\sin 8t \cos 4x$.
- $4u_{xx} = 4\big(-3\cos 2t \cos x + 2\sin 8t \cos 4x\big) = u_{tt}$.
- Check the data: $u(x,0) = 3\cos x$, $u_t(x,0) = 1 + 0 - \cos 4x$, $u_x(0,t) = u_x(\pi,t) = 0$.
- The term t comes from the eigenvalue 0: the average initial velocity $A_0^* = \frac{1}{\pi}\int_0^\pi g = 1$ makes the string drift upward.
- Since the data are finite cosine sums, no integrals were needed.



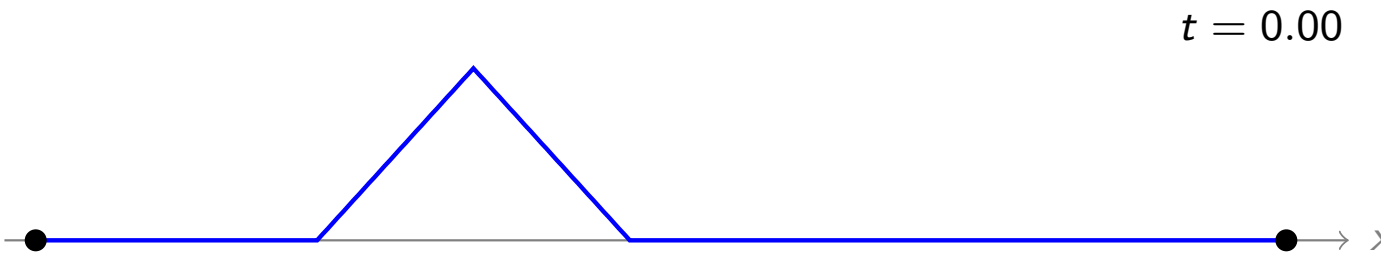
Picture: Reflection at Fixed and Free Ends



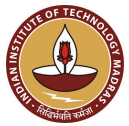
- A pulse released from rest ($k = 1$, $L = 4$) at times $t = 0, 1.2, 1.6, 2.6$.
- Fixed end: the reflected pulse is inverted. Free end: it returns upright (doubling at the end).



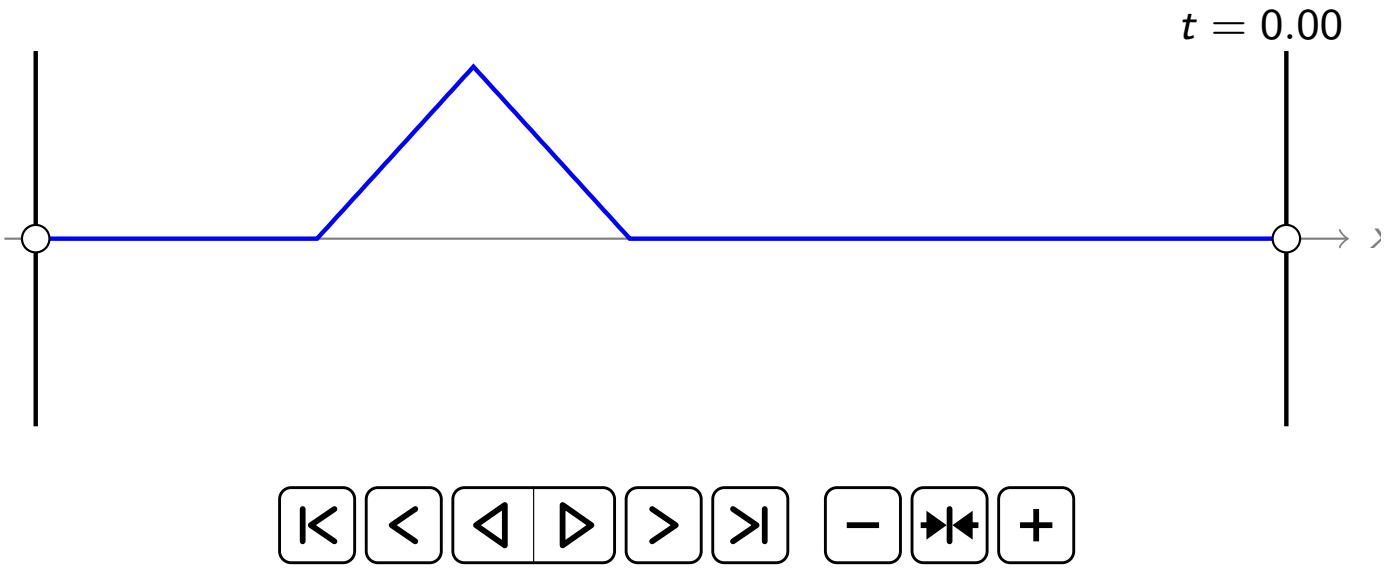
Animation: Pulse on a String with Fixed Ends



- $u(0,t) = u(L,t) = 0$: odd periodic extension; the pulse is inverted at every reflection.
- After time $2L/k$ the string returns to its initial shape.



Animation: Pulse on a String with Free Ends



- $u_x(0,t) = u_x(L,t) = 0$: even periodic extension; the ends (rings) slide up and down.
- The pulse is reflected without inversion at each free end.

Mixed Boundary Conditions

- Example 10. Solve $u_{tt} - u_{xx} = 0$, $0 < x < \pi$, $t > 0$, with

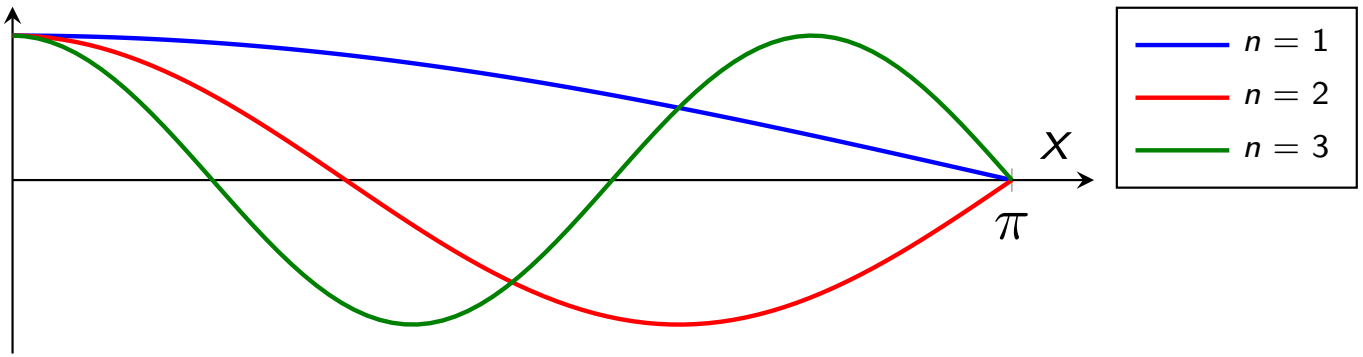
$$u_x(0,t) = 0, \quad u(\pi,t) = 0, \qquad u(x,0) = \cos\frac{5x}{2}, \quad u_t(x,0) = x^2.$$

- Free at $x = 0$ and fixed at $x = \pi$. With $u = XT$: $X'' - \mu X = 0$, $T'' - \mu T = 0$, $X'(0) = 0 = X(\pi)$.
- $\mu = 0$: $X = c_1 x + c_2$; $X'(0) = c_1 = 0$, $X(\pi) = c_2 = 0$. Trivial.
- $\mu = p^2 > 0$: $X = c_3 e^{px} + c_4 e^{-px}$; $X'(0) = 0 \Rightarrow c_3 = c_4$; $X(\pi) = c_3(e^{p\pi} + e^{-p\pi}) = 0 \Rightarrow c_3 = 0$. Trivial.
- $\mu = -p^2 < 0$: $X = c_1 \cos px + c_2 \sin px$; $X'(0) = c_2 p = 0 \Rightarrow c_2 = 0$.
- $X(\pi) = c_1 \cos p\pi = 0 \Rightarrow p = \dfrac{2n-1}{2}$, $n \in \mathbb{N}$.

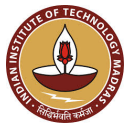


Example 10 (contd.): Eigenfunctions

- Eigenfunctions: $X_n(x) = \cos \dfrac{(2n-1)x}{2}, \quad n \in \mathbb{N}.$



- Zero slope at the free end $x = 0$, zero value at the fixed end $x = \pi$: quarter-wave modes.
- Time factors: $T_n = F_n \cos \dfrac{(2n-1)t}{2} + G_n \sin \dfrac{(2n-1)t}{2}.$



Example 10 (contd.): Coefficients

- By superposition,

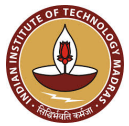
$$u(x,t) = \sum_{n=1}^{\infty} \left[F_n \cos \frac{(2n-1)t}{2} + G_n \sin \frac{(2n-1)t}{2} \right] \cos \frac{(2n-1)x}{2}.$$

- $u(x,0) = \sum F_n \cos \frac{(2n-1)x}{2} = \cos \frac{5x}{2}$: $F_3 = 1$ and $F_n = 0$ for $n \neq 3$.
- $u_t(x,0) = \sum \frac{2n-1}{2} G_n \cos \frac{(2n-1)x}{2} = x^2$, so

$$\frac{2n-1}{2} G_n = \frac{2}{\pi} \int_0^{\pi} x^2 \cos \frac{(2n-1)x}{2} \, dx.$$

- With $m = \frac{2n-1}{2}$: $\sin m\pi = (-1)^{n+1}$, $\cos m\pi = 0$, and

$$\int_0^{\pi} x^2 \cos mx \, dx = (-1)^{n+1} \left(\frac{\pi^2}{m} - \frac{2}{m^3} \right).$$



Example 10 (contd.): Solution

- Hence

$$G_n = \frac{2}{\pi m}(-1)^{n+1}\left(\frac{\pi^2}{m} - \frac{2}{m^3}\right) = \frac{8}{\pi}(-1)^{n+1}\left[\frac{\pi^2}{(2n-1)^2} - \frac{8}{(2n-1)^4}\right].$$

- The solution is

$$u(x,t) = \cos\frac{5t}{2}\cos\frac{5x}{2} + \frac{8}{\pi}\sum_{n=1}^{\infty}(-1)^{n+1}\left[\frac{\pi^2}{(2n-1)^2} - \frac{8}{(2n-1)^4}\right] \times \sin\frac{(2n-1)t}{2}\cos\frac{(2n-1)x}{2}.$$

- Check: at $t = 0$ the series vanishes and $u = \cos\frac{5x}{2}$; each term satisfies both end conditions.



Summary: Boundary Conditions and Eigenfunctions

- For $u_{tt} = k^2 u_{xx}$ on $0 < x < L$:

Ends	Conditions	Eigenfunctions	Eigenvalues p_n
fixed–fixed	$u = 0,\ u = 0$	$\sin \frac{n\pi x}{L},\ n \geq 1$	$\frac{n\pi}{L}$
free–free	$u_x = 0,\ u_x = 0$	$\cos \frac{n\pi x}{L},\ n \geq 0$	$\frac{n\pi}{L}$
free–fixed	$u_x = 0,\ u = 0$	$\cos \frac{(2n-1)\pi x}{2L},\ n \geq 1$	$\frac{(2n-1)\pi}{2L}$
fixed–free	$u = 0,\ u_x = 0$	$\sin \frac{(2n-1)\pi x}{2L},\ n \geq 1$	$\frac{(2n-1)\pi}{2L}$

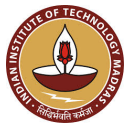
- Time factors: $\cos(kp_n t)$ and $\sin(kp_n t)$; for $p_0 = 0$ use $A_0 + A_0^* t$.
- Coefficients: expand f and g in the eigenfunctions using orthogonality.
- Fixed ends reflect waves inverted; free ends reflect them upright.



Homework: Finite String

- (1) $u_{tt} = k^2 u_{xx}$, $u(0,t) = u(L,t) = 0$, $u(x,0) = \sin\frac{\pi x}{L} + 3\sin\frac{4\pi x}{L}$, $u_t(x,0) = 0$.
- (2) $u_{tt} = 4u_{xx}$, $0 < x < \pi$, fixed ends, $u(x,0) = 0$, $u_t(x,0) = \sin^3 x$.
- (3) $u_{tt} = k^2 u_{xx}$, fixed ends, $u(x,0) = x(L-x)$, $u_t(x,0) = 0$.

  Ans: $u = \dfrac{8L^2}{\pi^3} \displaystyle\sum_{n \text{ odd}} \frac{1}{n^3} \cos\frac{n\pi kt}{L} \sin\frac{n\pi x}{L}$.
- (4) Plucked string with $b = L/2$: show only odd harmonics occur, and write u explicitly.
- (5) $u_{tt} = u_{xx}$, $0 < x < 1$, fixed ends, $u(x,0) = 0$, $u_t(x,0) = 1$.
- (6) $u_{tt} = k^2 u_{xx}$, free ends, $u(x,0) = \cos^2\frac{\pi x}{L}$, $u_t(x,0) = 0$.



Homework: Free and Mixed Ends

- (7) $u_{tt} - 25u_{xx} = 0,\ 0 < x < 5,\ t > 0;\ \ u(0,t) = 0,\ u_x(5,t) = 0;$
$$u(x,0) = \sin 7x, \quad u_t(x,0) = \begin{cases} 1, & 0 < x < 2, \\ x, & 2 \le x < 5. \end{cases}$$
- (8) $u_{tt} = u_{xx},\ 0 < x < \pi,\ u_x(0,t) = u_x(\pi,t) = 0,\ u(x,0) = x,$ $u_t(x,0) = 0.$
- (9) Find the eigenvalues and eigenfunctions for $u(0,t) = 0,$ $u_x(L,t) = 0,$ and write the general solution.
- (10) Show that $E(t) = \frac{1}{2}\int_0^L (u_t^2 + k^2 u_x^2)\,dx$ is constant for fixed or free ends (conservation of energy).
- (11) Use energy to show that the fixed-end problem has at most one solution.
- (12) Verify that the series solution with $g = 0$ equals $\frac{1}{2}[f^*(x-kt) + f^*(x+kt)]$ for $f(x) = \sin\frac{\pi x}{L}$.

